Backtesting Williams %R Strategy

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Introduction

Background

Williams %R, developed by renowned trader Larry Williams in the 1970s, is a momentum-based technical analysis indicator that measures overbought and oversold conditions in the market to identify potential reversal points in asset prices (Gratton). It gained popularity across various markets, including stocks, commodities, and futures, during a period when technical analysis was becoming increasingly significant in financial trading. Williams is also known for his influential writings, such as "How I Made One Million Dollars Last Year Trading Commodities," where he discusses the development and application of Williams %R among other trading strategies (Gratton).

Strategy Overview

A Williams %R strategy involves using the Williams %R indicator to identify overbought and oversold conditions in the financial market. These conditions can signal potential reversal points. The basic strategy revolves around choosing a lookback period, setting overbought/oversold Williams %R indicator bounds, and taking a long/short position accordingly (Groette). In this case, the reference article suggests that a Williams %R indicator strategy only performs well when it is used to take long positions rather than short positions. Furthermore, the authors also claim this strategy can be boosted by combining it with other indicators such as the RSI (Groette).

Strategy Description

Indicators

This Williams %R strategy revolves around using the Williams %R indicator which is an oscillating momentum indicator that reflects the current close relative to the "highest high" for a given lookback period (Groette). This indicator ranges from -100 to 0 with values closer to 0 indicating overbought conditions and values closer to -100 indicating oversold conditions (Gratton). Commonly, the lookback period used for this indicator is 14 days and the overbought threshold is above -20 and the oversold threshold is below -80. However, in the strategy I am recreating, the authors performed hyperparameter tuning on a lookback period of 2-25 days and various threshold bounds (Groette). Therefore, in this analysis, I will be using a lookback period of 2 days, an oversold threshold of -90, and an overbought threshold of -30.

The formula for the Williams %R indicator is:
(highest high(last x days) – close) / (highest high(last x days) – lowest low(last x days))* -100

Where x is in this instance is the lookback period.

Buy and Sell Signals

In this strategy, we enter a long position when the Williams %R indicator crosses below -90, signifying the asset is recovering from oversold conditions. We exit the long position when the Williams %R indicator crosses above -30 or when the close is higher than yesterday's high.

Additional Consideration

The article mentions that they combined this strategy with an RSI indicator for stronger results and that the Williams %R indicator performs poorly for "shorts" (Groette). This motivates

next steps that I will discuss at the end of the analysis. For the purposes of this report, I will only be reviewing my results from backtesting this basic Williams %R strategy.

Backtesting

Backtesting Framework

A disclaimer for this report: I chose to implement this strategy on my own without a Python framework. I had previously used the `WilliamsR` indicator function available in the backtrader python framework. However, the results from using that predefined function were largely different from what the authors of this report claim. Therefore, I implemented the logic of this indicator, strategy, and metrics independent of a set framework. I used Python's MatPlotLib library to generate plots.

Assumptions

We use the SPDR S&P 500 ETF Trust or SPYfor short from its inception on January 29th 1993 until the date the article is posted June 14th, 2024 (Groette). This data was scraped from yahoo finance at the granularity level of daily. Each row of the data table contains the date, daily high, daily low, close, adjusted close, and volume. The article does not directly mention what granularity of data they use to achieve those results. However, based on the strategy formula we assume that the data used was daily (Groette). Furthermore, the article does not mention a starting principal amount - in our analysis we assume both $100,000 and $125,000. This article does not mention backtesting with leverage - as such we also do not incorporate leverage into my methodology. The article does not mention taking into account a commission percentage for trades. We explore strategy performance across different commission percentages to evaluate how robust the strategy is to those different percentages.

Metrics

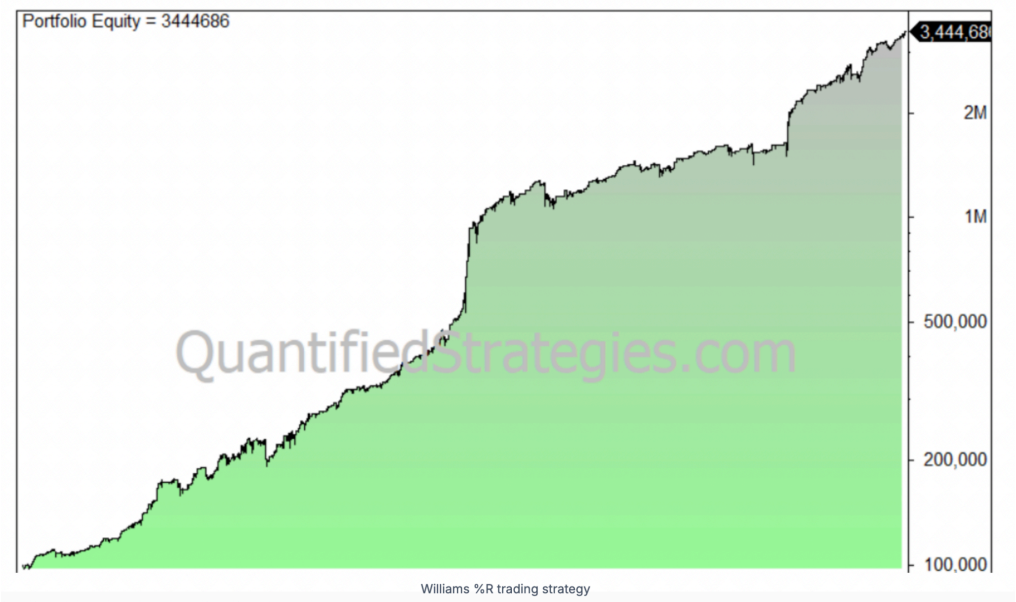
The metrics we used were Compound Annual Growth Rate (CAGR) and Maximum Drawdown (Max Drawdown).

CAGR is a metric that represents the mean annual growth rate of an investment over a specified period, assuming profits are reinvested at the end of each period. It provides a smoothed rate of return, eliminating the effects of volatility during the time frame. This makes it a useful measure for comparing the performance of investments with different time horizons (Eichler).

Max Drawdown measures the largest peak-to-trough decline in the value of an investment over a given period. It is expressed as a percentage and indicates the maximum potential loss an investor could have experienced during the period. Max Drawdown is a key metric for assessing downside risk, as it highlights the worst-case scenario for an investment's performance (Corvin).

Original Strategy and Results

The authors of the original strategy outlined above only specify the lookahead period, thresholds used, and an approximate time frame for SPY on which backtesting was performed. The equity curve below illustrates the portfolio's alleged performance as reported by the author. This equity curve suggests the principal investment was $100,000. However, the author did not explicitly disclose the initial principal investment, and the plot lacks proper labeling. Notably, there is no label for the x-axis, and the dollar values on the y-axis are inconsistent with the tick marks. This plot is the sole piece of "'evidence" provided in the article, raising concerns about the credibility and potential misrepresentation of the results. Accompanying the plot, the authors also claim this strategy achieves a Max Drawdown of 17% and CAGR of 11.5% with 580 trades over the period of 21 years (Groette).



Methods

Our initial approach involved testing the strategy as presented in the article and comparing its performance to a simple buy-and-hold strategy over the same time frame. However, after observing discrepancies in the portfolio values reported by the authors, we attempted to replicate their results by optimizing the initial investment, increasing the lot amount, and adding leverage (which were not explicitly provided). Additionally, we examined how the results varied by adjusting the commission rate.

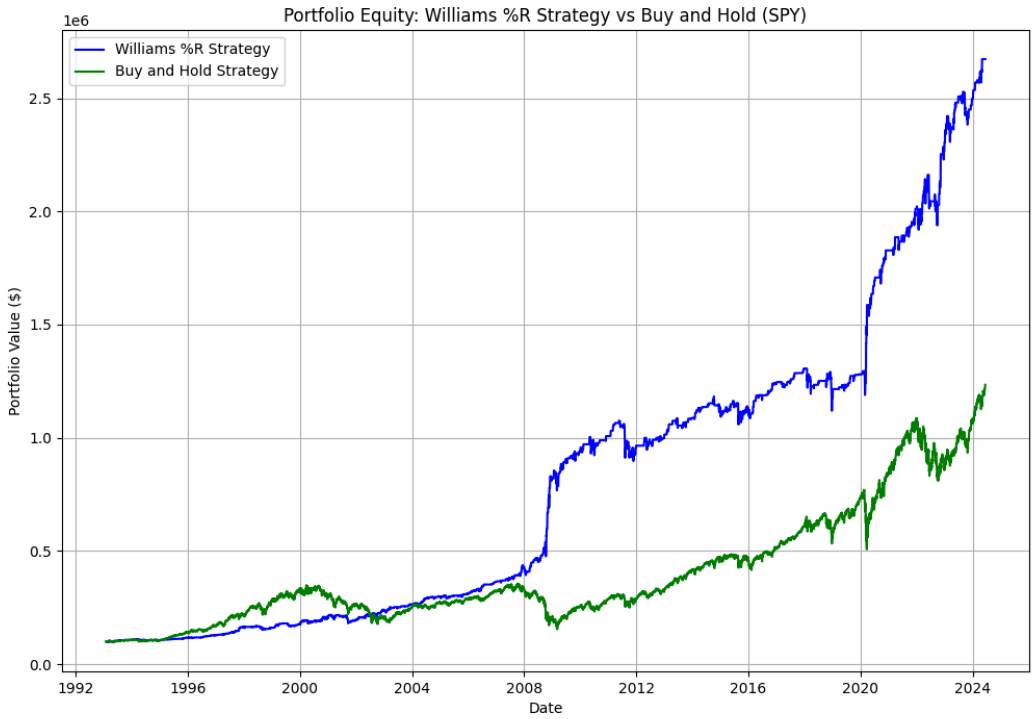
Results

Initial Results

The figure below contains the equity curve of the strategy detailed in the article vs. buying and holding this ETF. To generate the results below, we assume a 0% commission on

trades. The table below contains the metrics the article used vs. the buy and hold. It appears that the CAGR, Max Drawdowns, and Number of Trades is approximately consistent with what was reported. However, while the shape of the two equity curves align, the total portfolio equity from this test is shy of $800,000. That being said, however, from these results, it appears that this Williams %R strategy has outperformed buying and holding across all metrics. Notably, the portfolio value of Williams %R is over 2x greater than that of buying and holding. The CAGR is slightly higher and the Max Drawdown is considerably lower. We also see from this plot, that during both the 2008 and 2020 recessions, this indicator greatly outperformed purely buying and holding.

	Williams %r	Buy and Hold
Portfolio Value	$2,672.705.52	$1,234,115.97
CAGR	11.04%	8.34%
Max Drawdown	17.55%	56.47%
# trades	600	1

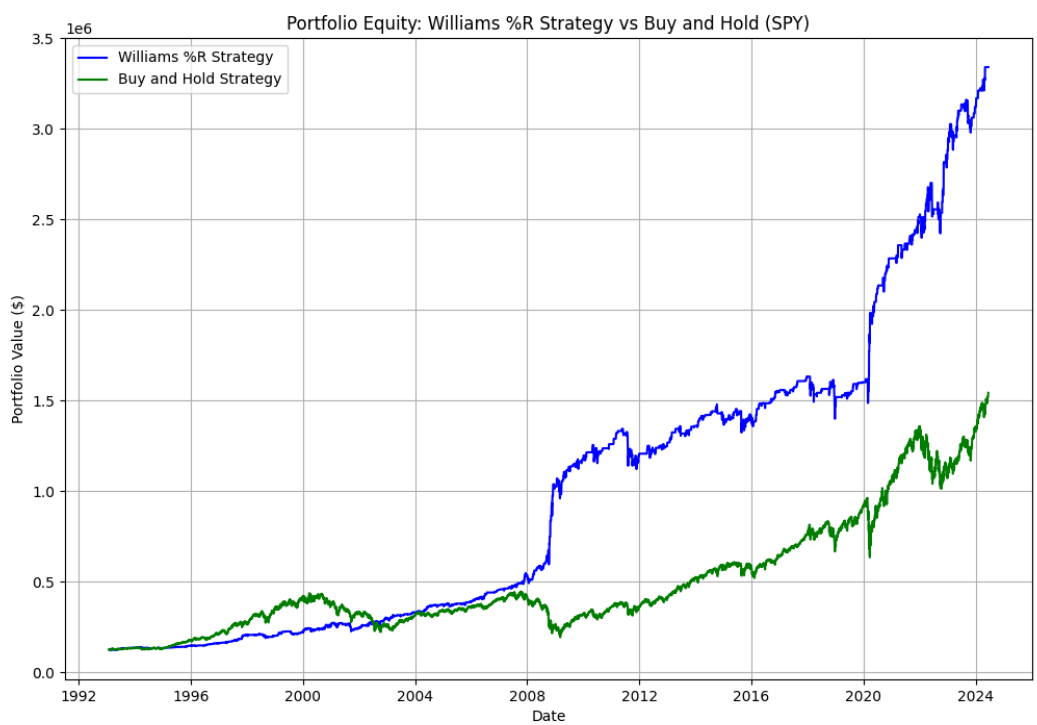


Replicating Portfolio Value

Initial Cash Investment

We attempt to replicate the original portfolio value by adjusting the assumed principal investment, increasing it to $125,000 to achieve results comparable to the author's findings. The metrics table and plot below show the strategy applied over the same time period, with this revised principal amount. The CAGR, Max Drawdown, and number of trades remain nearly identical to the previous results. The only notable change is that the total portfolio value now approaches the reported $3.4 million. This suggests that the actual principal investment used by the authors was higher than implied in the original plot.

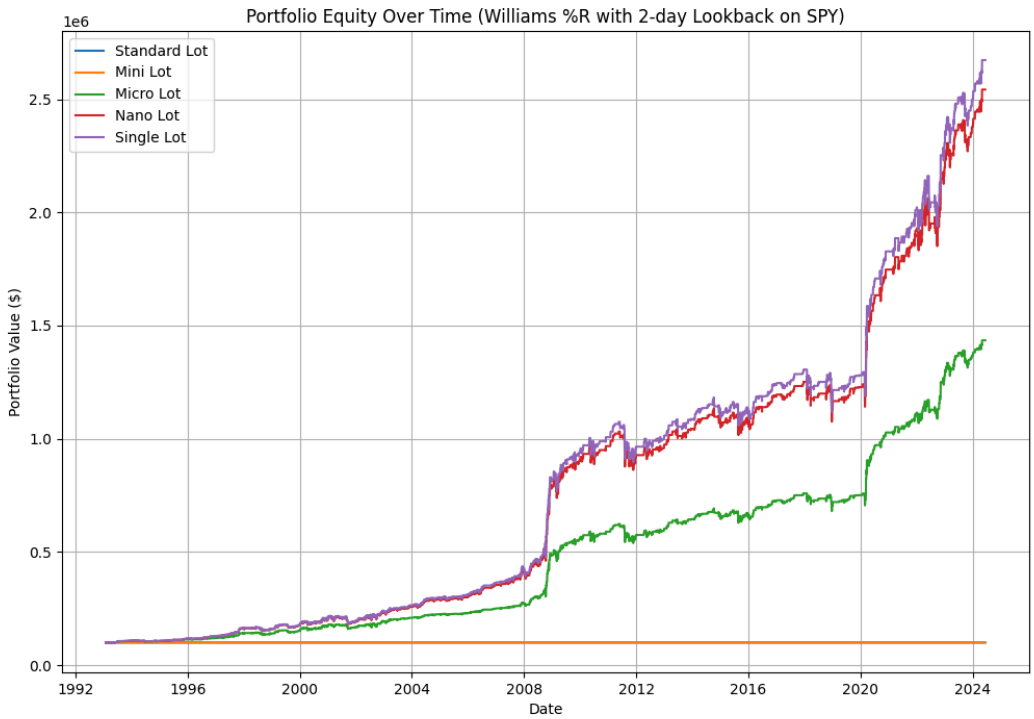
	Williams %r	Buy and Hold
Portfolio Value	$3,341,323.86	$1,542,769.58
CAGR	11.04%	8.34%
Max Drawdown	17.55%	56.47%
# trades	600	1



Lot Size

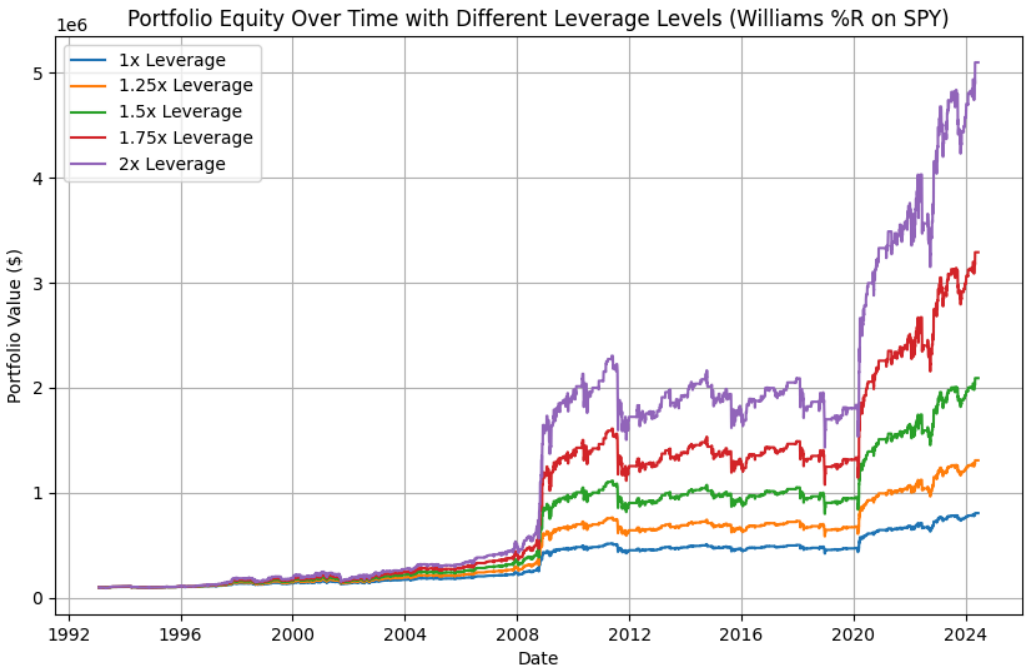
Next, we try to replicate the original portfolio using various lot sizes, which refer to the standardized number of shares traded in a single transaction. So far, we've been using a lot size of 1. However, we experiment with different sizes—100,000 (standard), 10,000 (mini), 1,000

(micro), 100 (nano), and 1 (single)—to determine if adjusting the lot size can boost returns and match the author's original portfolio results. As shown in the plot, smaller lot sizes correspond to higher Williams %R returns. However, once the lot size drops to 100 or below, returns plateau at just over $2.5 million, suggesting the authors did not achieve their gains by simply increasing the lot size.



Leverage

Finally, we increase leverage to replicate the author's initial portfolio equity. We find that the portfolio reaches the same total equity when leverage is raised by 75%, to 1.75x. However, any leverage below this level fails to achieve the same results. If the authors used leverage to enhance their backtesting outcomes, they did not disclose it. This lack of transparency is misleading, as they present the Williams %R strategy as inherently "successful" when, in reality, it may be dependent on additional cash flow to achieve those results.

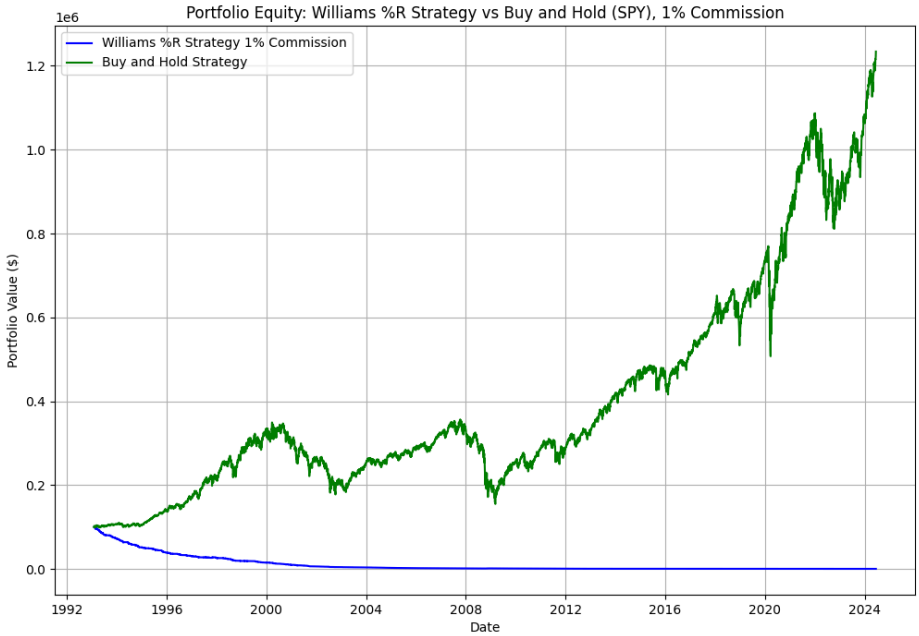


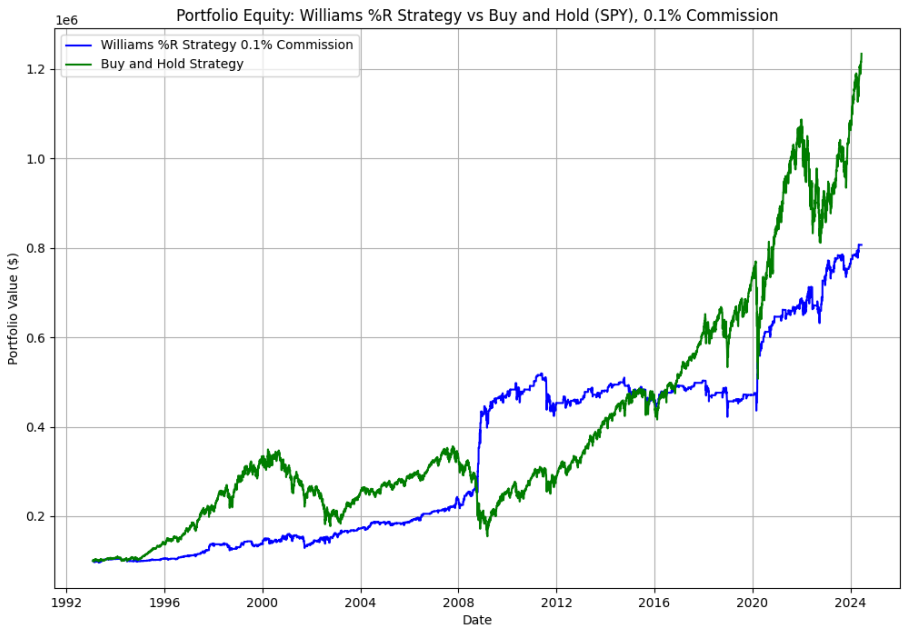
Changing Commission Rate

We also investigate how changing the commission rate affects the results of this trade by replicating the previous procedure using a principal of $100,000 but with a 0.1% and a 1% commission. The plots and table of metrics below show the difference in portfolio equity with varying commission rates. From these results, we can see that this strategy is not robust to different commission rates and even a commission rate as low as 0.1% diminishes returns from this strategy. At a commission rate of 1%, buy and hold exponentially outperforms this Williams %R strategy.

	Williams %r 0%	Williams %r 0.1%	Williams %r 1%	Buy and Hold
Portfolio Value	$2,672,705.52	$806,628.41	$196.90	$1,542,769.58

CAGR	11.04%	6.88%	-18.01%	8.34%
Max Drawdown	17.55%	19.84%	99.8%	56.47%
# trades	600	600	651	1





Discussion

This research revealed the importance of clear reporting and transparency in trading strategies. The lack of key details, such as the initial investment, made it challenging to replicate the original results accurately. Unexpectedly, increasing the principal investment to $125,000 closely matched the reported portfolio value, raising concerns about the validity of the original findings. Additionally, we found the strategy's performance was worse than a simple buy-and-hold approach in terms of risk (Max Drawdown) and return (CAGR) when adding a commission. This further suggests that the original article is misleading about the effectiveness of the strategy. The key takeaway is that thorough scrutiny of assumptions and parameters is essential before adopting a strategy. Since extraordinary claims require extraordinary evidence, Further analysis is definitely needed, including de-trending the data and ensuring that there is no lookahead bias. This will provide a more complete understanding of the strategy's actual performance and risks before drawing conclusions.

Sources

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